Presentation Script

A Unified Knowledge-Grounded Generative AI Framework

[Slide 1: Title Slide]

Script:

> *"Good morning, respected members of the panel and my fellow peers. I am **Souparna Paul**, an M.Tech candidate in the Department of Computer Science and Engineering. Today, I am honored to present my research titled: **'A Unified Knowledge-Grounded Generative AI Framework for Adaptive Dialogue Summarization and Intelligent Information Retrieval.'** This project is conducted under the esteemed guidance of **Dr. Soma Chatterjee**."*

[Slide 2: Abstract / Executive Summary]

Script:

> *"The core vision of this research is to bridge the fundamental gap between **linguistic fluency and factual accuracy**. While Generative AI has revolutionized content creation, it often lacks a 'grounded' truth.*
>
> *This work is an ambitious extension of my previous research—specifically my work on **'AutoMeet++'** for meeting summarization and the **'LLM Doctor'** RAG system. We are moving toward a grander framework that doesn't just summarize based on internal patterns but dynamically verifies every insight against a validated external knowledge base to **eliminate hallucinations**."*

[Slide 3: Introduction & Context]

Script:

> *"In the current digital landscape, we deal with massive amounts of conversational data. Summarization has evolved from simple **'Extractive'** copy-pasting to **'Abstractive'** rewriting.*
>
> *However, dialogues are inherently messy—full of slang, speaker interruptions, and implicit references. To handle this, we introduce **'Knowledge Grounding'**—the process of anchoring the AI's output to a verified source of truth, ensuring that the model remains contextually aware and factually reliable."*

[Slide 4: Problem Statement]

Script:

> *"We have identified three critical pain points:*
>
> 1. ***'Semantic Drift'**: LLMs often lose the thread in conversations exceeding 20 turns.*
> 2. ***'Fact-Checking Gap'**: LLMs prioritize grammar over truth, leading to plausible lies or hallucinations.*
> 3. ***'Stale Knowledge'**: Traditional models cannot summarize events occurring after their training cutoff.*
>
> *Our goal is simple: **How can we ensure a dialogue summary is 100% factually grounded while remaining concise?**"*

[Slide 5: Literature Review & Gap Analysis]

Script:

> *"In my recent systematic review of RAG architectures, we analyzed models like **BART, T5, and GPT-4**. We found two major gaps:*
>
> - *Current models treat dialogues as isolated text blocks, ignoring the 'outside world.'*
> - *Evaluation metrics like **ROUGE** focus on word overlap rather than factual correctness.*
>
> *This research addresses the lack of unified frameworks that combine **Dense Vector Retrieval** with the specific nuances of multi-party conversations."*

[Slide 6: Research Objectives]

Script:

> *"My objectives are four-fold:*
>
> 1. *To design a **modular RAG-based pipeline** for real-time semantic search.*
> 2. *To implement a **'Filtering Layer'** that handles the noise and filler words typical in human speech.*
> 3. *To develop an **'Adaptive Retrieval'** mechanism that decides when the model needs external data versus when it should rely on the dialogue itself.*
> 4. *To establish a **rigorous benchmarking system** using both linguistic and factual consistency metrics."*

[Slide 7: Proposed Methodology]

Script:

> *"The technical core involves three stages:*
>
> - ***First: 'Input Processing'** for speaker identification.*
> - ***Second: The 'Retrieval Engine'** using FAISS or ChromaDB for dense passage retrieval.*
> - ***Finally: 'Generation'** using Llama-3 or Mistral.*
>
> *We are specifically utilizing **LoRA (Low-Rank Adaptation)** to fine-tune these models for specialized domains without the massive computational overhead of full retraining."*

[Slide 8: System Architecture Overview]

Script:

> *"Here we see the architecture: The dialogue enters a **'Fusion Layer'** where it meets retrieved context from our Knowledge Base. We use a **Bi-Encoder** for embeddings and a **Transformer decoder** for generation.*
>
> *Crucially, we've added a **'Verifier'**—a secondary factual consistency check that scores the output against the retrieved evidence before the user sees it."*

[Slide 9: Datasets & Evaluation Framework]

Script:

> *"For training and validation, we are using **DialogSum** for general conversations and **MediaSum** for high-density factual data from news interviews.*
>
> *We are moving beyond ROUGE scores; our evaluation includes **'FactCC'** using Natural Language Inference models to measure entity precision and factual faithfulness."*

[Slide 10: Anticipated Novelty & Contribution]

Script:

> *"The novelty lies in **'Dynamic Contextualization.'** This isn't a static summarizer; it's a unified pipeline merging Information Retrieval and Generative AI. This is a significant step toward **trustworthy AI** in sensitive domains like Legal or Finance, where a single hallucination can have serious consequences."*

[Slide 11: Conclusion & Next Steps]

Script:

> *"Currently, I have successfully completed the environment setup and initial data indexing.*
>
> ***My immediate next steps involve:***
>
> - *Initiating domain-specific LoRA fine-tuning.*
> - *Refining the noise reduction layer.*
>
> ***Looking further ahead:***
> *I aim to extend this to **Multimodal Retrieval** for video calls and optimize it for edge devices using **GGUF quantization**.*
>
> *Thank you for your time. I am now open to any questions."*